

MUHAMMAD HARIS HUSSAIN

Artificial Intelligence Engineer

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PROFESSIONAL SUMMARY

Results-driven AI Engineer and final-year BS Artificial Intelligence student at SZABIST with hands-on experience building production-grade machine learning systems, deep learning models, and full-stack AI-integrated web applications. Proficient in PyTorch, TensorFlow, Hugging Face Transformers, and scikit-learn, with applied expertise in NLP and computer vision architectures including ViT, DeBERTa, and BERT. Demonstrated ability to architect, train, deploy, and optimize AI pipelines for real-world use cases including deepfake detection, healthcare AI, and smart traffic management. Passionate about leveraging AI to solve complex engineering challenges.

TECHNICAL SKILLS

AI / ML	PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, ViT, DeBERTa, BERT, OpenCV, YOLOv8, NLP, Computer Vision, Deep Learning, Reinforcement Learning
Languages	Python, Java, HTML5, PowerShell, Lua
Frameworks	Flask, FastAPI, React, Streamlit
Databases	MySQL, SQLite, Firebase
Cloud & DevOps	AWS, GitHub, Anaconda, OpenAPI
Hardware	Arduino, Raspberry Pi, Cisco (Networking)
Tools	Jupyter, VS Code, Postman, Twilio, GPT API

PROJECTS

FakeDetect — AI-Powered Deepfake & Generated-Text Detection System	2024 – 2025
Tech Stack: Python · Flask · React · Hugging Face Transformers · ViT · DeBERTa-v3-large · SQLite · JWT · PyTorch	
- Designed and implemented a full-stack web application for detecting AI-generated images (deepfakes) and AI-written text, achieving high-confidence classification using Vision Transformer (ViT) and DeBERTa-v3-large models loaded via Hugging Face Transformers. - Engineered REST API endpoints in Flask with JWT-based user authentication, SQLite history storage, and an admin panel for analytics and user management. - Applied Test-Time Augmentation (TTA) and confidence threshold calibration to improve model robustness and reduce false positives in image detection. - Built a responsive React frontend with heatmap visualization, analysis history panel, mobile-optimized UI, and real-time progress indicators.	
AI-Based Smart Traffic Management System	2024 – 2025
Tech Stack: Python · FastAPI · React · TypeScript · YOLOv8 · PostgreSQL · Docker · CI/CD	
Developed an end-to-end AI traffic management platform with YOLOv8-based real-time vehicle detection, congestion prediction ML models, and adaptive signal timing algorithms.	

- Built a FastAPI backend with 30+ RESTful endpoints, PostgreSQL schema design, and React/TypeScript/Tailwind frontend, following IEEE SRS documentation standards.
- Configured Docker containerisation and CI/CD pipelines for automated testing and deployment.

AI Hotel Management System

2024

Tech Stack: [Python](#) · [Flask](#) · [Machine Learning](#) · [MySQL](#)

- Engineered an AI-powered hospitality platform automating guest reservations, service recommendations, and business analytics using supervised learning models.
- Developed predictive occupancy models to optimise room allocation and revenue forecasting.

ViitalCare Custom GPT — Healthcare AI Assistant

2024

Tech Stack: [Python](#) · [OpenAI GPT API](#) · [Hugging Face](#) · [NLP](#) · [Prompt Engineering](#)

- Developed a domain-specific GPT assistant for healthcare, leveraging advanced prompt engineering and Hugging Face model integrations to provide accurate, safe medical information.

FixTrack — Smart Maintenance & Issue Tracking System

Jan 2024 – Jun 2024

Tech Stack: [Python](#) · [Firebase](#) · [Web App](#)

- Built a maintenance workflow platform enabling real-time issue logging, task assignment, priority management, and resolution tracking for facilities teams.

EDUCATION

Bachelor of Science in Artificial Intelligence

2023 – Present

SZABIST University, Karachi • Relevant: Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures

Diploma of Associate Engineer in Software Technology

2021 – 2023

Aligarh Institute of Technology

ACHIEVEMENTS & AWARDS

- ZABEFEST 2024 Runner-Up — Competed in the annual tech competition at SZABIST University, placing runner-up among multiple teams, demonstrating strong problem-solving and applied AI skills.
- Capstone AI Role — Selected as AI Model Integration lead for the 3rd-year Fake Detection project, responsible for training, optimising, and deploying ViT and DeBERTa-v3-large deep learning models via Hugging Face Transformers.

CORE COMPETENCIES

Machine Learning Engineering • Deep Learning Architecture • NLP & Computer Vision • Hugging Face Transformers • REST API Development • Full-Stack Integration • Agile / Team Collaboration • Technical Documentation